Training and validation curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Training and validation curves before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learning rate experiments

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Training a small network

Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Keep the restricted network h = ReLU(x), y = vh, with only v trained. **Training** pairs remain (1, 2) and (2, 4). Start v = 0 and make four full-data updates at rate 0.1. Save states v = 0, 1, 1.5, 1.75 and 1.875.

At each saved state, evaluate both splits without changing v during evaluation.

For this deliberately constructed demonstration, validation pairs are (3, 4.5) and (4, 6). Their targets follow a different slope from the **training** pairs.

This makes the two trends disagree visibly; it is not **evidence** that these few rows represent a real population. Validation labels do not enter the gradient.

Updates	v	Training MSE	Validation MSE
0	0	10	28.125
1	1	2.5	3.125
2	1.5	0.625	0
3	1.75	0.15625	0.78125
4	1.875	0.0390625	1.7578125

Both columns use ordinary MSE in the same target units, evaluated on their own fixed rows. **Training** MSE is 2.5(v - 2) squared; validation MSE is 12.5(v - 1.5) squared.

Different **input** values and target rules affect the loss levels. A gap alone does not identify its cause.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Training and validation curves trace and record the first step you can explain without help.

Use W2: Use the declared rule for **Training** and **validation curve**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Training and validation curves explanation against the supplied facts before discussing it.

Use the worked case for **Training** and **validation curve**s. Proposed explanation: “The last **checkpoint** is always best”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Evaluate new checkpoints independently

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Recover a plotted point

> Recover a plotted point. Apply the supplied Training and validation curves facts and keep your reasoning.

At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Use the declared rule

> Use the declared rule. Apply the supplied Training and validation curves facts and keep your reasoning.

Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

My reasoning and evidence

W3 Repair incompatible measures

> Repair incompatible measures. Apply the supplied Training and validation curves facts and keep your reasoning.

A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Evaluate new checkpoints independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Evaluate new checkpoints independently. Apply the supplied Training and validation curves facts and keep your reasoning.

Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

Explain the disagreement with selection by **training** loss.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Limit a curve diagnosis

> Limit a curve diagnosis. Apply the supplied Training and validation curves facts and keep your reasoning.

Training loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Training and validation curves to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **training** and **validation curve**s **evidence** sheet to Image-pattern Network Investigation. Show how you can compute comparable **training** and validation losses at saved **checkpoint**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Training and validation curves

> Plot training and validation loss for checkpoints before opening final cases. Explain the labelled step before checking the changed condition.

Plot **training** and validation loss for **checkpoint**s before opening final cases.

Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/15

Checkpoint losses

	training	validation
epoch 1	4	3
epoch 2	2	2
epoch 3	1	2.5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/15

Training and validation curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read training curve

Training loss falls 4 -> 2 -> 1.



2. Read validation curve

Validation loss falls 3 -> 2 then rises to2.5.



3. Choose checkpoint

Select epoch 2, not epoch 3, under the declared validation criterion.

CHANGE ONE THING

Epoch 3 validation loss is instead 1.5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plot **training** and validation loss for **checkpoint**s before opening final cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Select epoch 2, not epoch 3, under the declared validation criterion.

1. Read **training curve**

2. Choose **checkpoint**

3. Read **validation curve**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose **checkpoint** Select epoch 2, not epoch 3, under the declared validation criterion. The **training curve** measures fit to examples used for learning.

The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen. Under the supplied **selection rule**, keep **checkpoint** two.

A later final test must remain outside both learning and **checkpoint** selection.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Epoch 3 validation loss is instead 1.5. Which conclusion is supported by the supplied facts?

1. Select epoch 3 under the unchanged criterion.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Select epoch 3 under the unchanged criterion. Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Repair a missing building block

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

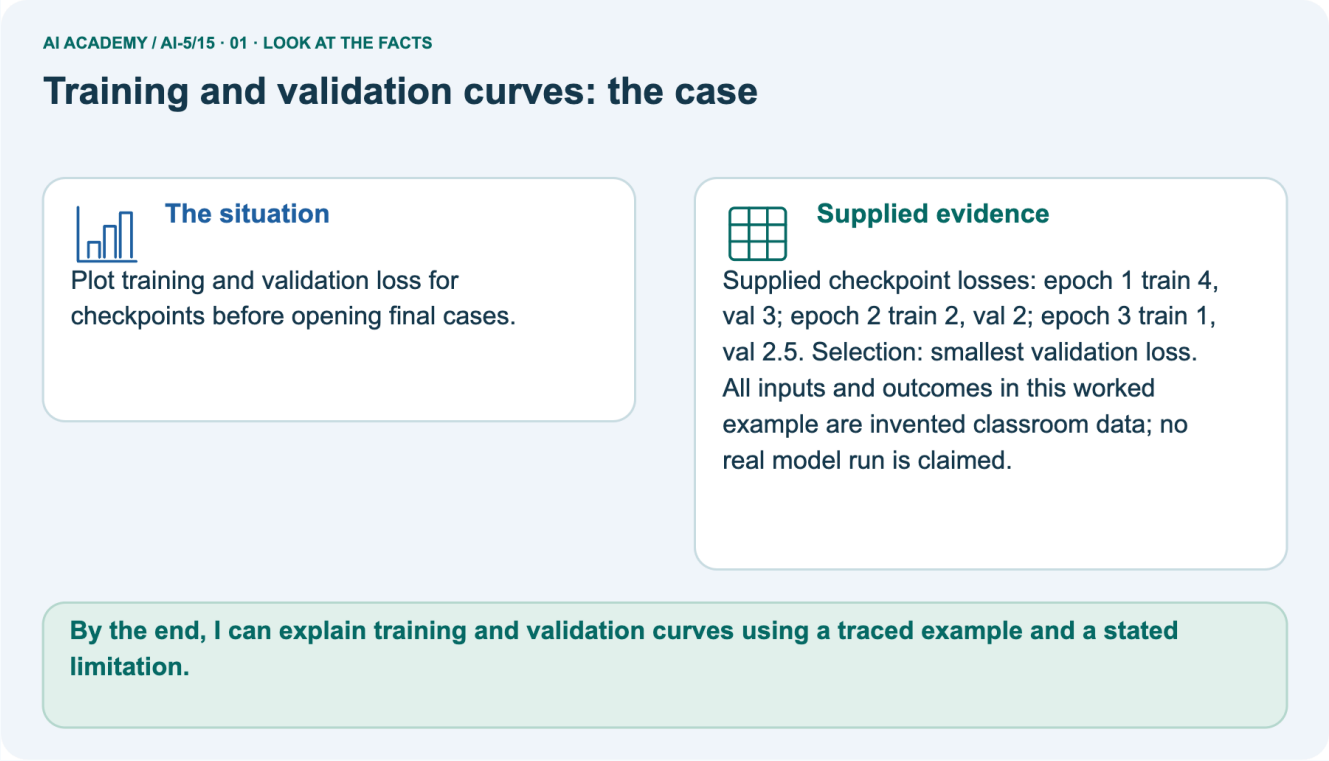
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Training and validation curves picture from its labels, then record a relevant limit.

01. Training and validation curves: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

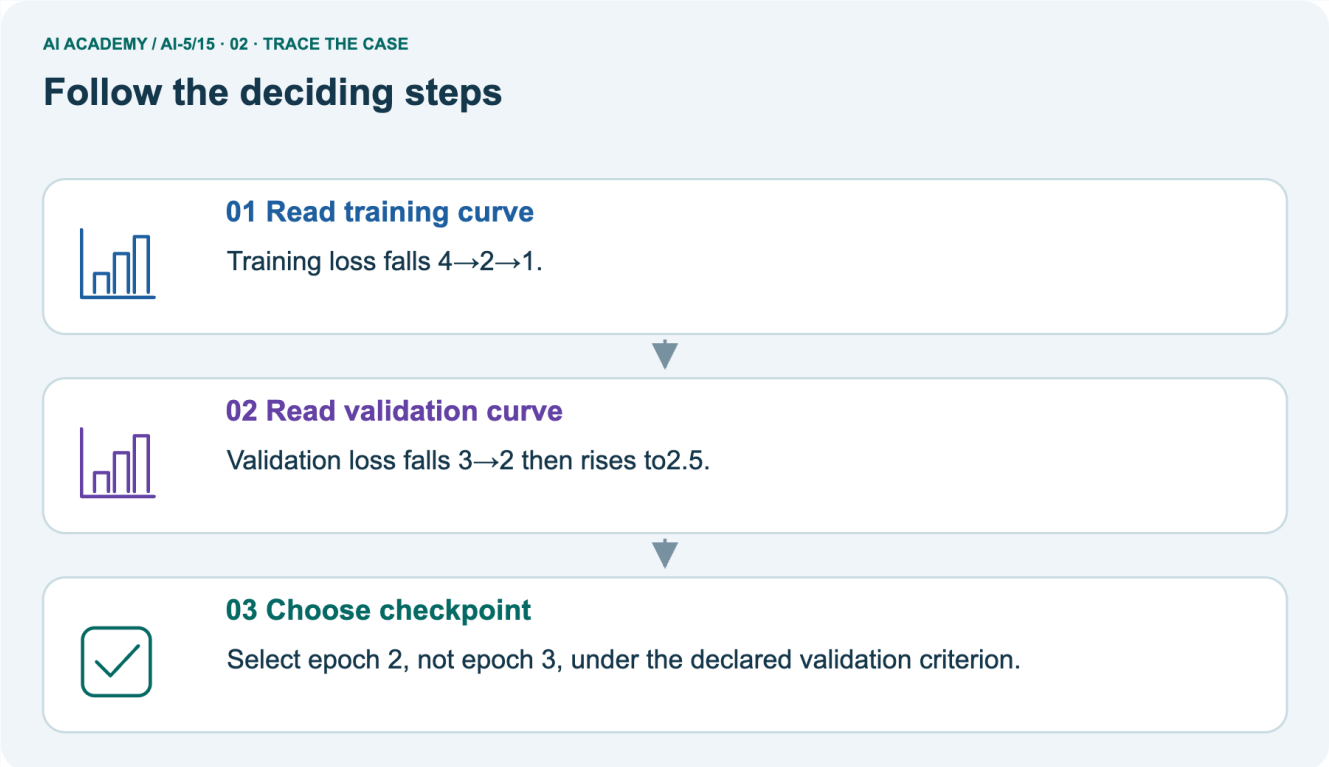
Plot **training** and validation loss for **checkpoint**s before opening final cases. Supplied facts: Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **training curve**: **Training** loss falls 4→2→1. → Read **validation curve**: Validation loss falls 3→2 then rises to2.5. → Choose **checkpoint**: Select epoch 2, not epoch 3, under the declared validation criterion.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Checkpoint losses



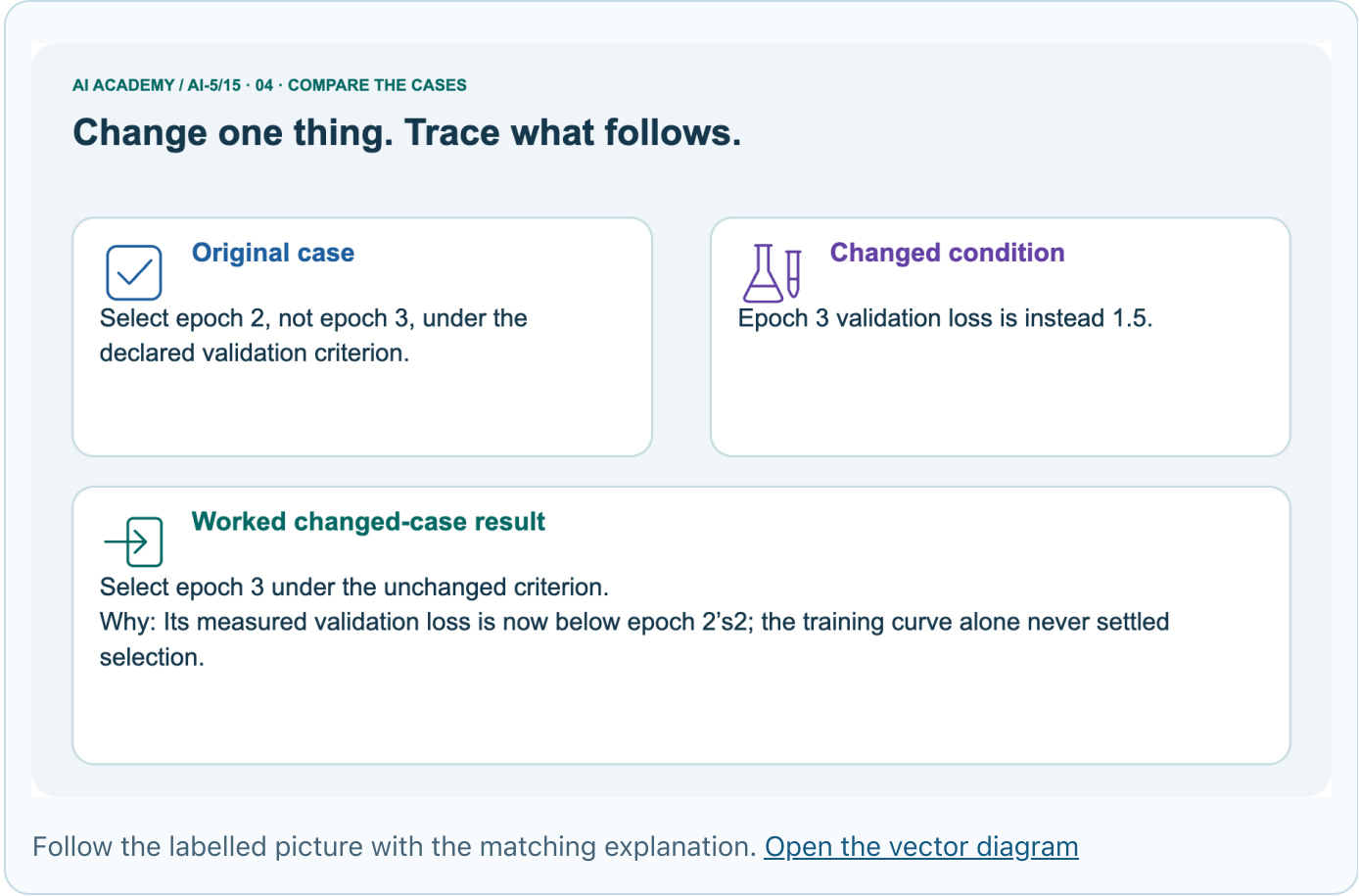
Follow the labelled picture with the matching explanation. Open the vector diagram

Checkpoint losses. epoch 1: **training** = 4, validation = 3. epoch 2: **training** = 2, validation = 2. epoch 3: **training** = 1, validation = 2.5.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

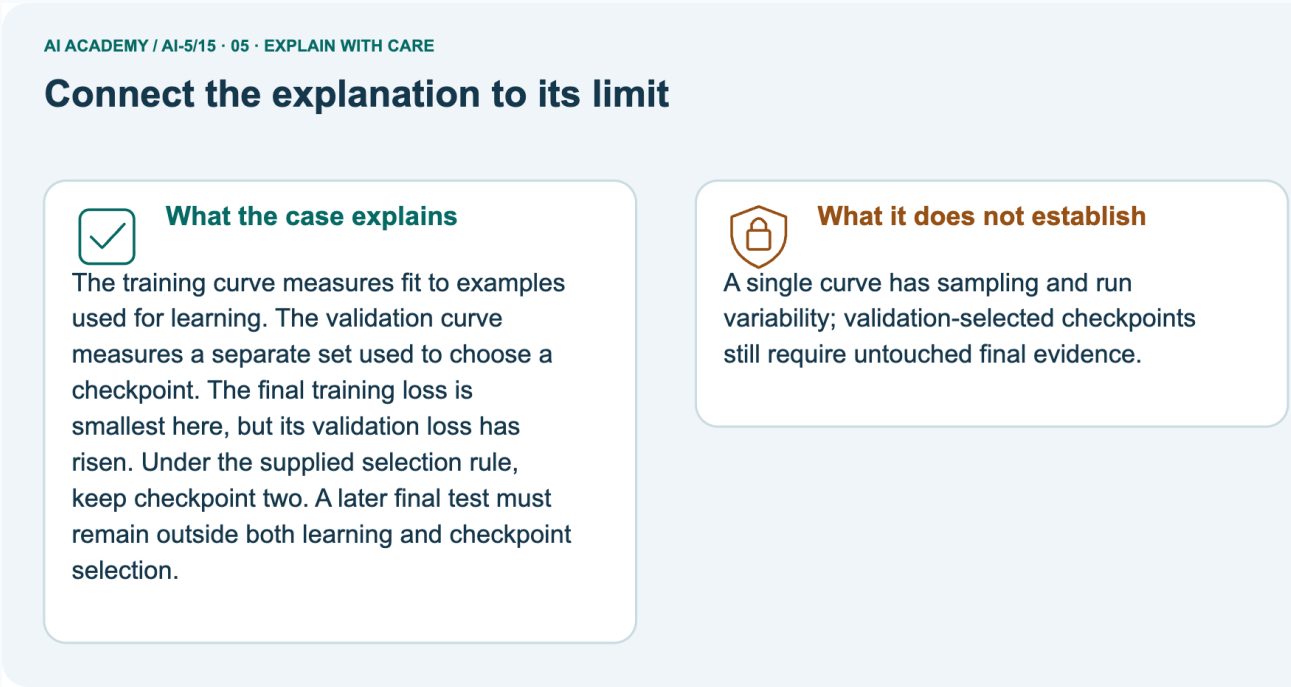
Original case: Select epoch 2, not epoch 3, under the declared validation criterion. Changed condition: Epoch 3 validation loss is instead 1.5. Expected result: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

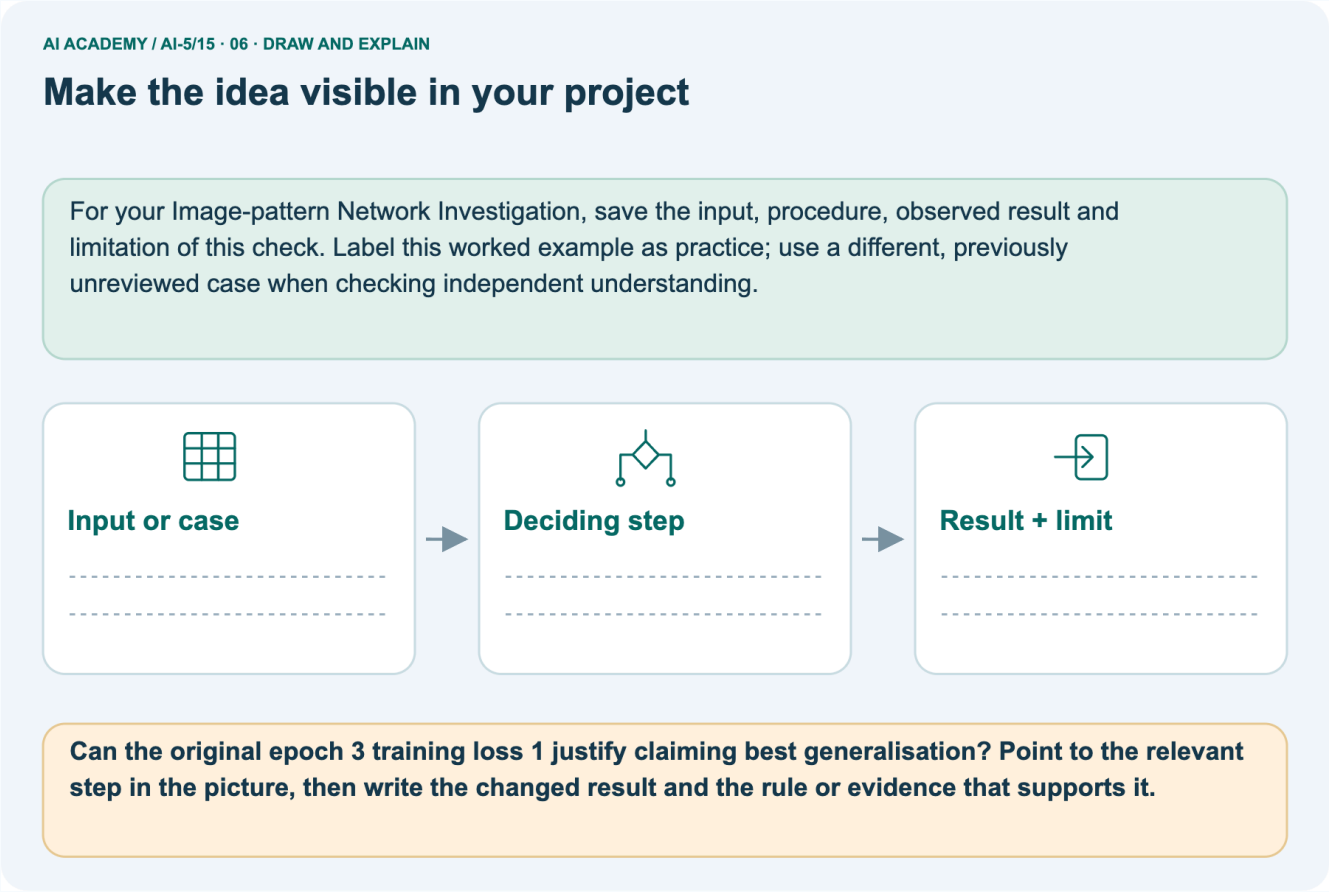
Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

Limit: A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.